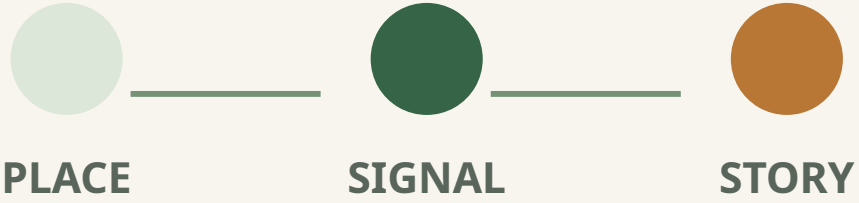
Earth Sensors Lab

A six-week accessible STEAM pilot



A discussion draft for ITP / IMA and high-school students

A shared investigation of a real place, shaped by access from the first question.

Kemi Sijuwade-Ukadike · Small Systems Lab

For conversation with Shawn Van Every

A field lab for shared observation

How do we make environmental observations that more people can shape and understand?

Students start with a local outdoor site and a question they can investigate. They combine direct observation, a small sensor or manual measurement, and a clear way to represent what they find.

Built from existing work

Earth Sensors Lab
Accessible environmental sensing and STEAM learning

Accessible by Design
Access is part of how a system is conceived and built

CC Fest workshop
A tested pattern for making, describing, testing, and documenting access choices

A bounded question, a complete investigation

EXAMPLE QUESTION

Where does garden soil dry first, and what can our readings actually tell us?

01

Observe

Note the place, conditions, and question.

02

Measure

Use a sensor or a repeatable manual check.

03

Represent

Make a map, chart, or p5.js view with a text equivalent.

04

Record

Save a Schema Card and a short field note.

Six weeks of field inquiry

One site and one question give the group a shared thread from first observation to public explanation.

WEEK 1

Question + access path

Choose a site and frame a question people can investigate in different ways.

WEEK 2

Site + sensing plan

Decide what to observe, what to measure, and what a sensor cannot tell us.

WEEK 3

Prototype + test

Collect initial observations and readings; record conditions and access choices.

WEEK 4

Data + representation

Build a simple p5.js or paper map with a text equivalent and readable labels.

WEEK 5

Interpret + revise

Compare patterns, name uncertainty, and revise the explanation with peer feedback.

WEEK 6

Share + reflect

Present a field record and prototype; name the next question worth investigating.

Access shapes every part of the investigation

Participation	Speak, type, sketch, observe, work with a partner, or pass.
Controls	Offer labeled keyboard access; keep the visual canvas optional.
Descriptions	Describe sensor setups, images, maps, and changes in plain language.
Data display	Pair charts with text summaries or a table; use readable labels and contrast.
Permission	Make motion or camera-based interaction optional, stoppable, and off by default.
Take-home record	Use a Schema Card to capture the access rule, test, and revision.

A shared studio across age groups

The partnership can connect creative technology with questions students bring from their own environments.

ITP / IMA

Bring creative technology, prototyping, and design critique. Students could mentor, collaborate, or develop a shared prototype with the high-school group.

High-school partners

Bring observations, questions, and local knowledge. Students choose how to participate and help decide what counts as a useful record.

SHARED OUTCOME

A tested prototype and a clear field record that shows how students observed, interpreted, and revised their claim.

A light pilot, shaped with ITP / IMA

Start with one site, one question, and a small set of tools. Keep the footprint adjustable.

Site + school

Where can students work safely, and which high-school partner makes sense?

Format + roles

Where could six weekly sessions fit, and who should co-lead or mentor?

Tools + access

Which sensor setup is reliable, and what no-device participation paths are needed?

Resources

What support is needed for materials, transportation, accommodations, and educator time?

Existing work and starting points

EARTH SENSORS LAB

ukadike.github.io/earth-sensors-lab/

Accessible environmental sensing, garden observation, and STEAM learning.

ACCESSIBLE BY DESIGN

ukadike.github.io/accessible-by-design-prototyping/

Workshop materials and a playground for testing access rules in creative systems.

CC FEST, OCTOBER 2026

The workshop tested a small pattern: make or describe one garden object, name an access rule, test it, and take away a Schema Card.

Could we shape this into a six-week pilot?

I’d value your help finding the right fit for ITP / IMA and a high-school partner.

- Where could this sit within ITP / IMA?
- Who should help shape the high-school partnership?
- What site, schedule, and support would make a small pilot workable?

WHAT I BRING **ESL framing, accessible field methods, facilitation, and workshop materials.**

A first conversation can define the scope and identify who should join it.